Ib Valley Coalfield

🏭 Introduction

The **Ib Valley Coalfield** is one of the major coal-producing regions in India, located in Odisha. It is primarily operated by Mahanadi Coalfields Limited, a subsidiary of Coal India Limited.

The coalfield is known for its **large reserves of non-coking (thermal) coal**, which are widely used for power generation. It hosts several important opencast mining projects such as Lakhanpur, Samaleswari, and Lajkura, making it a key contributor to India's coal production. Ib Valley plays a vital role in supplying coal to thermal power plants in eastern and central India.

🌍 Geographical Overview – Ib Valley Coalfield

The Ib Valley Coalfield is located in the **Jharsuguda and Sundargarh districts** of Odisha and forms part of the **Mahanadi basin coalfields**.

📍 Location & Setting

- Situated along the **Ib River valley** in western Odisha
- Lies in an industrially important region with good connectivity
- Well connected by rail and road networks for coal transportation

🌄 Physical & Geological Features

- Part of the **Gondwana coalfields**, containing extensive coal deposits
- Features **thick and gently dipping coal seams**, suitable for opencast mining
- Coal type: **non-coking thermal coal**

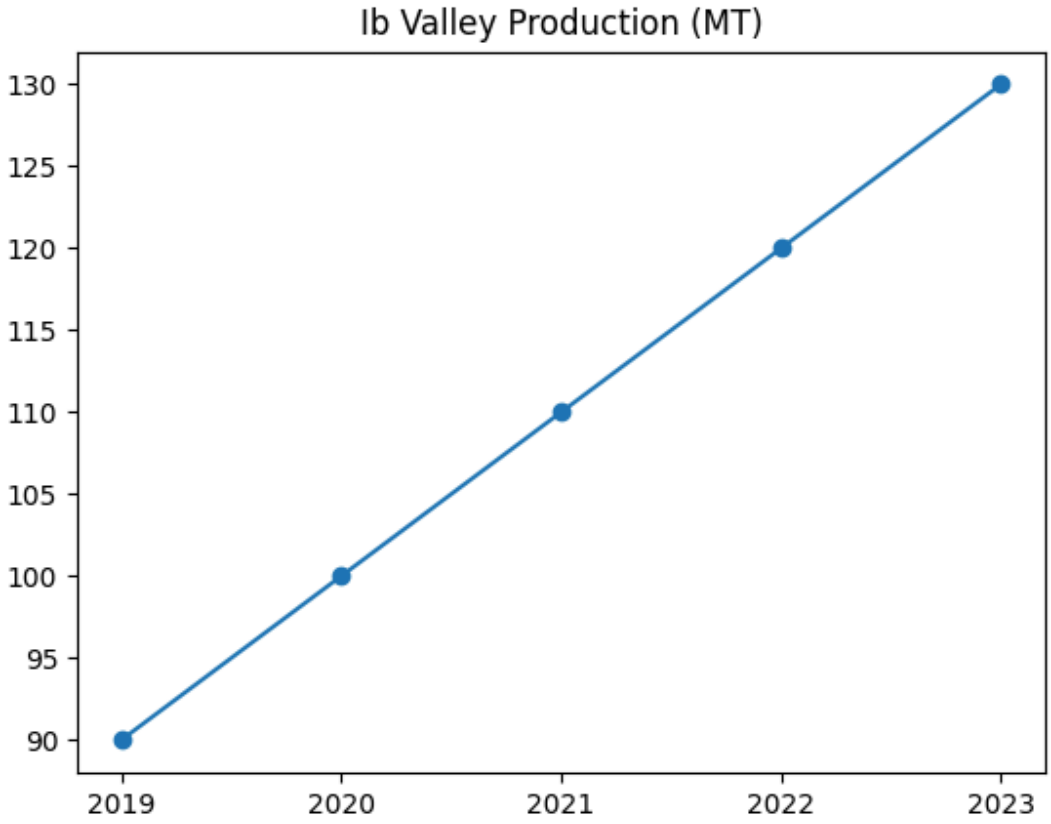
🌊 Nearby Natural Features

- Located along the Ib River, a tributary of the Mahanadi River
- Surrounded by industrial and power generation zones

🌦️ Climate

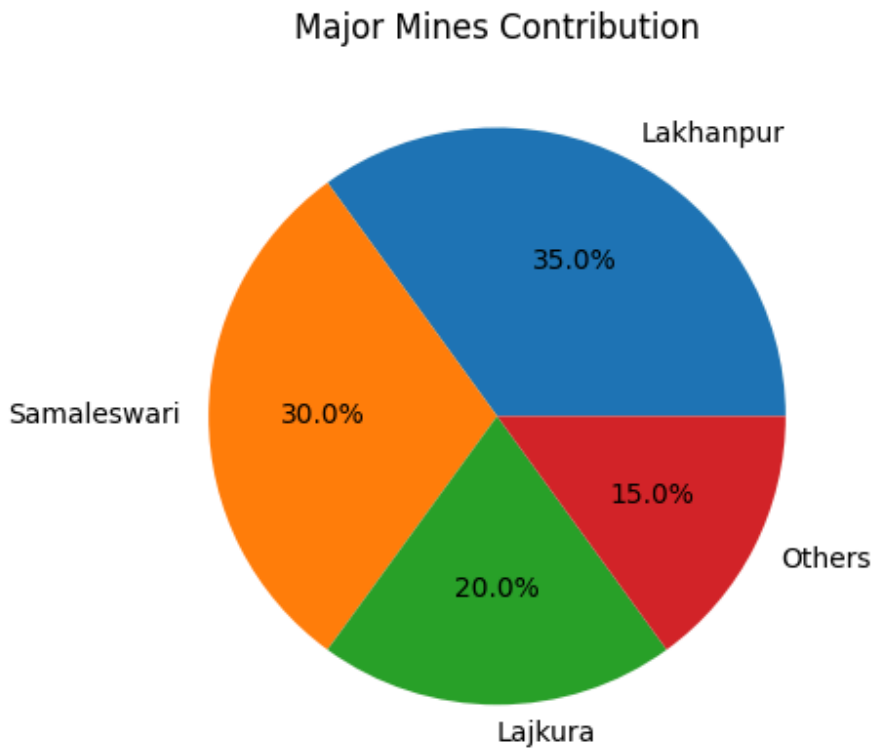
- Tropical climate
- Hot summers (up to ~45°C)
- Moderate to heavy monsoon rainfall
- Mild winters

📊 1. Production Trend (MT)



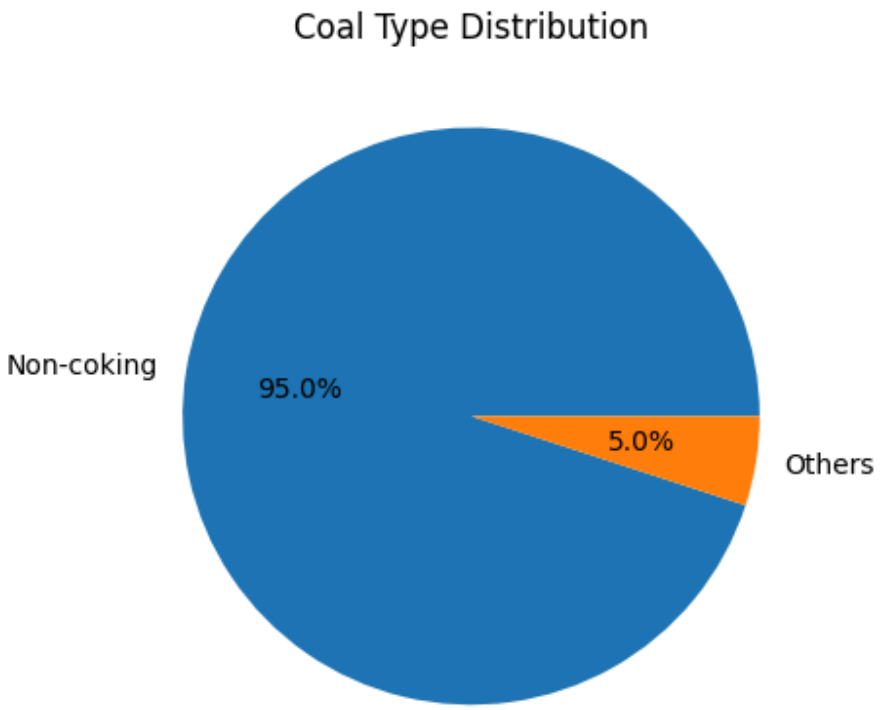
This chart shows the steady increase in coal production from Ib Valley under Mahanadi Coalfields Limited.
The growth reflects expansion of large opencast projects.
Significance: Useful for production forecasting and understanding regional contribution to India's coal output.

📊 2. Major Mines Contribution



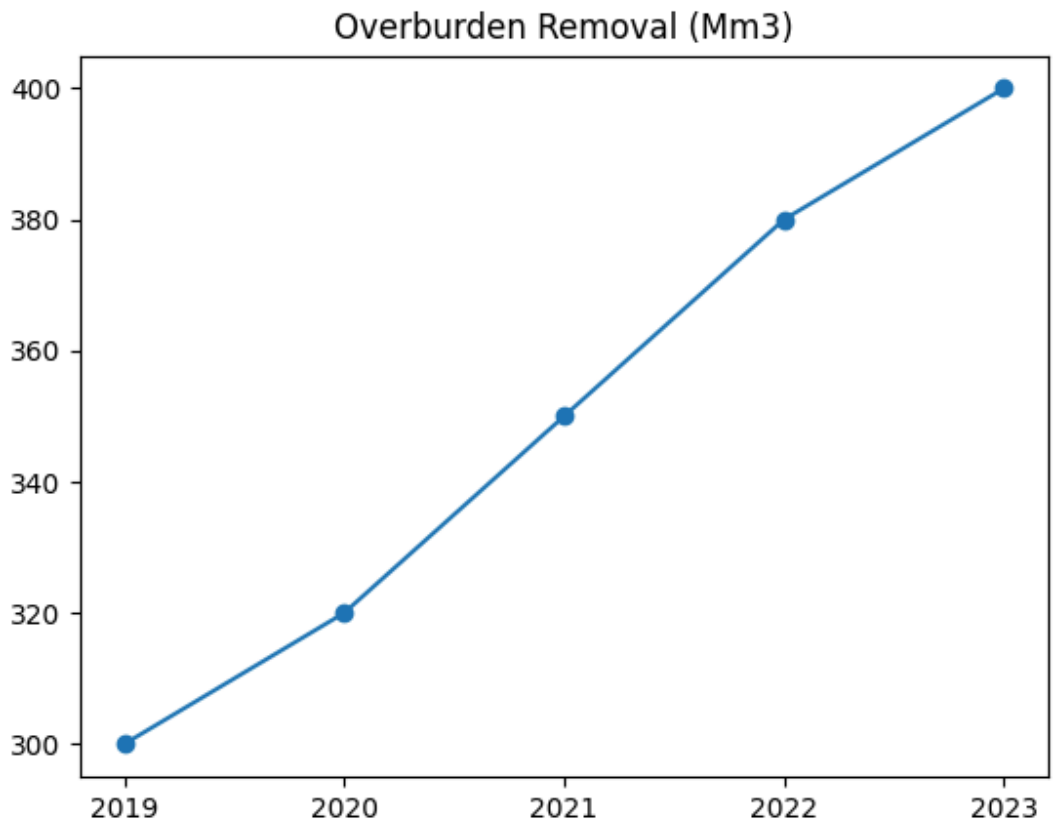
Represents the share of major mines like Lakhanpur, Samaleswari, and Lajkura.
Each mine contributes differently based on reserve size and capacity.
Significance: Helps identify key production centers and prioritize resource allocation.

📊 3. Coal Type Distribution



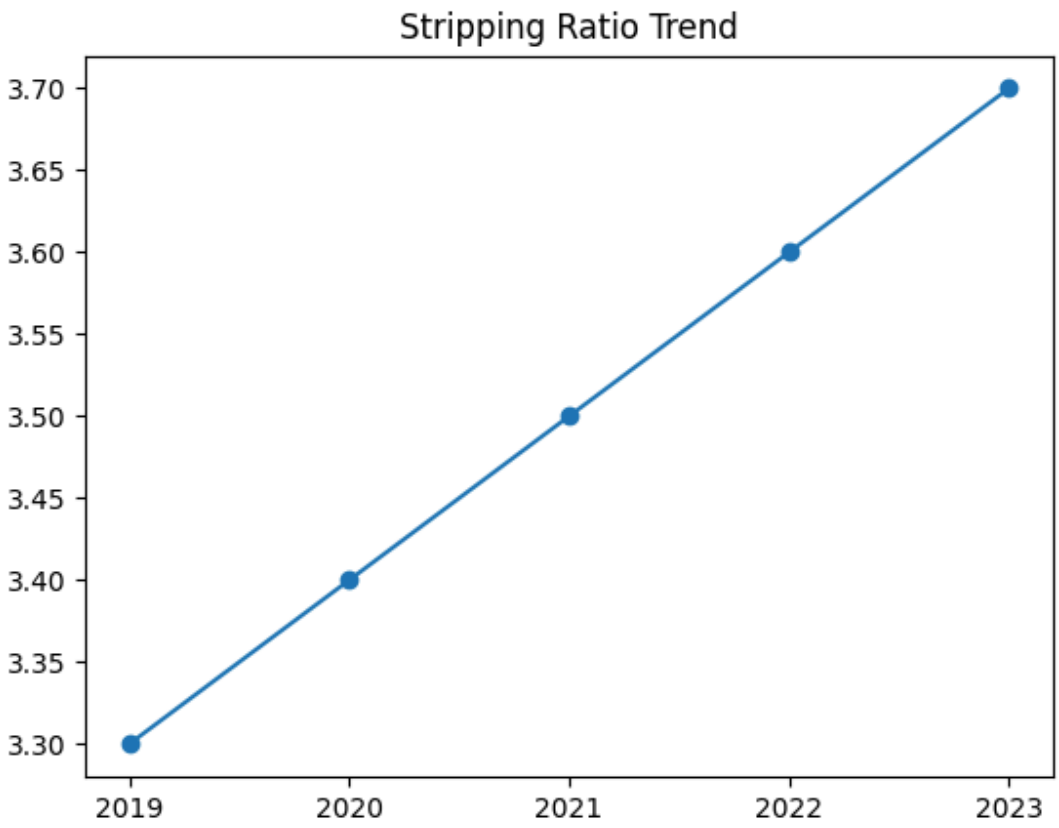
Shows dominance of non-coking (thermal) coal.
This coal is primarily used in power generation.
Significance: Determines end-use sector and highlights importance in electricity production.

📊 4. Overburden Removal (Mm³)



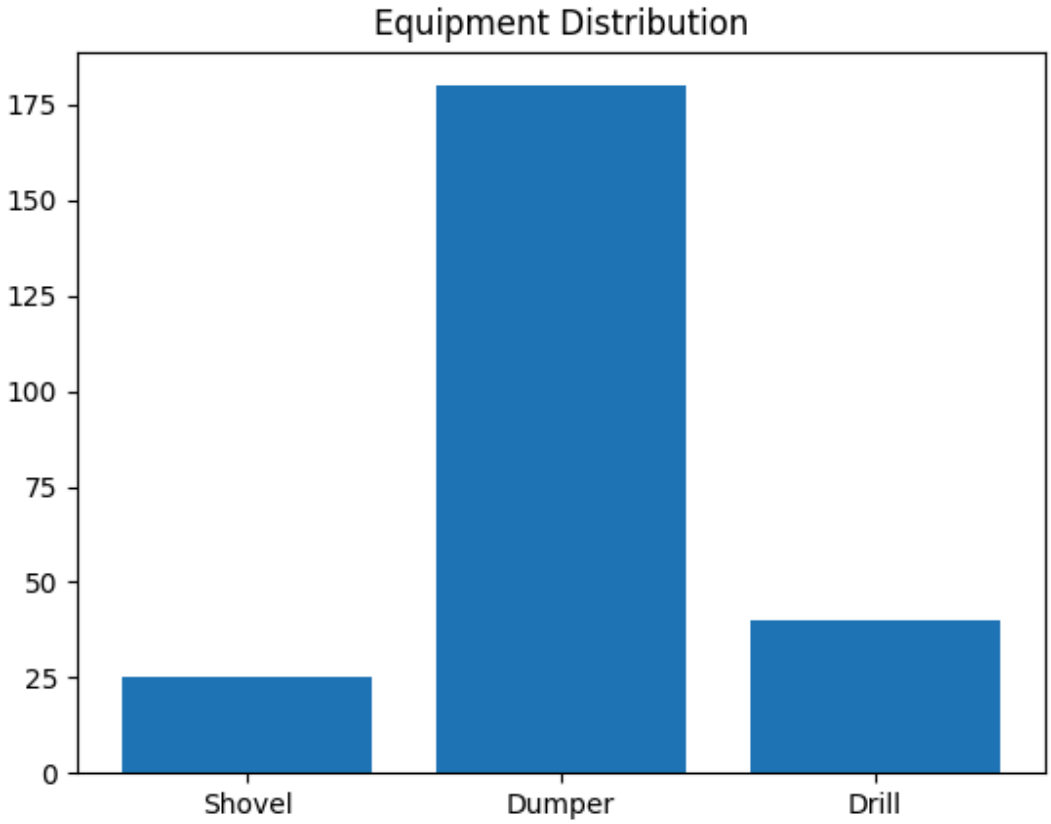
Displays the volume of overburden removed annually.
Increasing values indicate expansion and deeper mining operations.
Significance: Important for dump design, equipment planning, and cost estimation.

📊 5. Stripping Ratio Trend



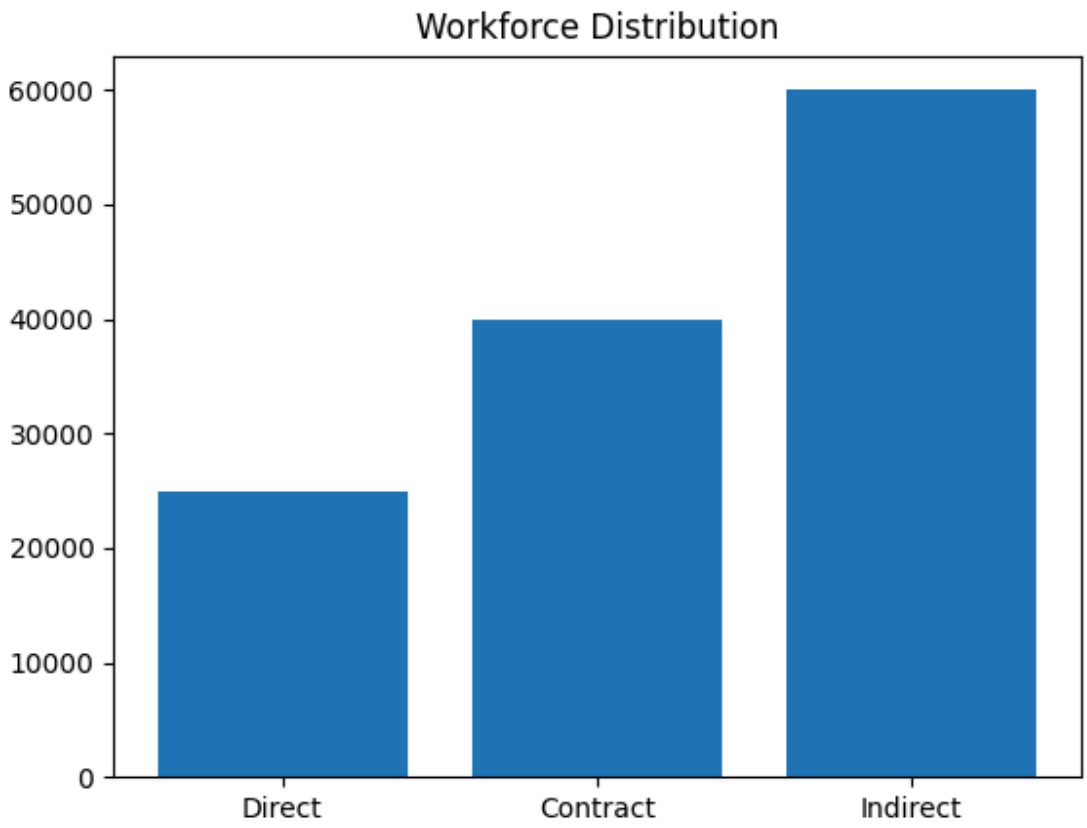
Shows the relationship between overburden and coal production.
A gradual increase indicates rising mining depth and cost.
Significance: Key parameter for economic feasibility and mine planning.

📊 6. Equipment Distribution



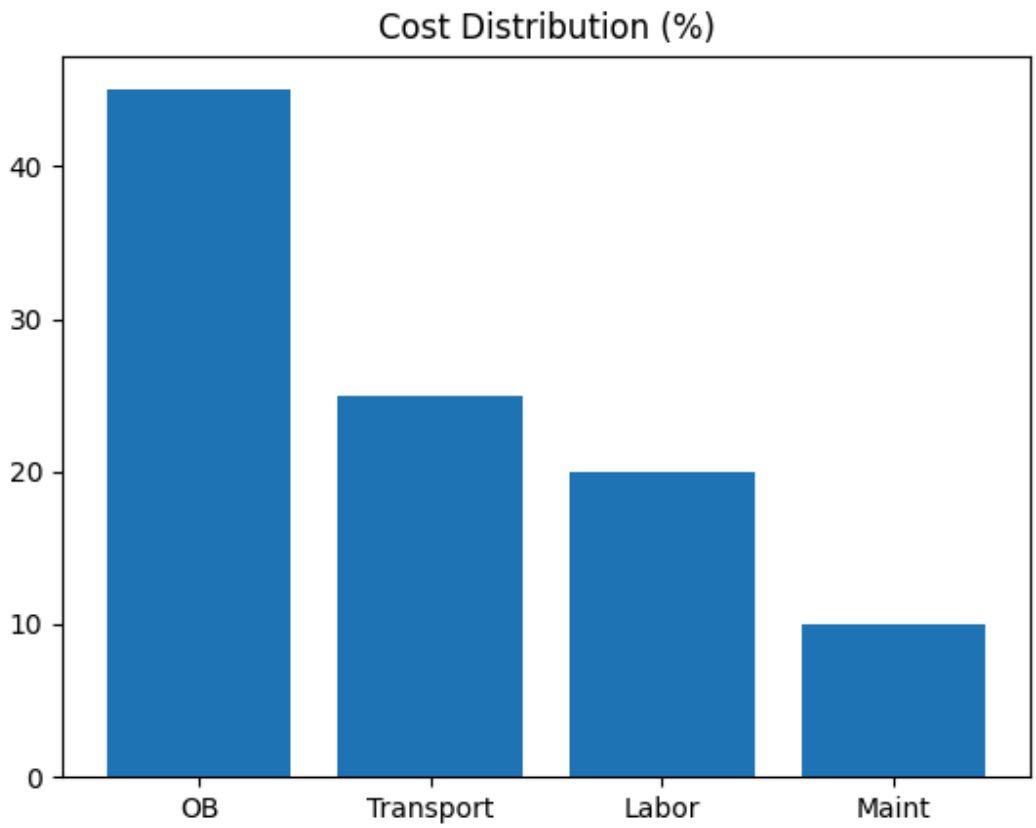
Represents deployment of major equipment like shovels, dumpers, and drills.
Dumpers dominate due to large-scale material handling.
Significance: Useful for fleet management and productivity optimization.

📊 7. Workforce Distribution



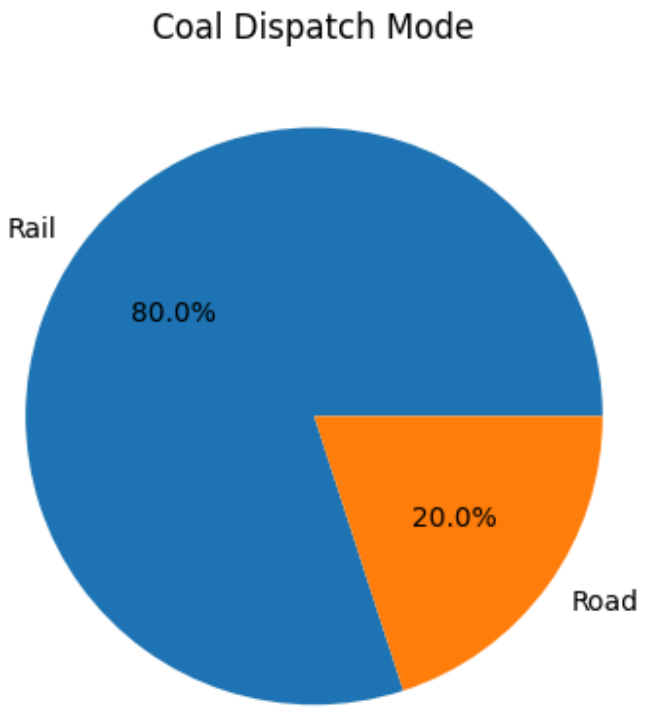
Displays manpower distribution across different employment categories.
Mining generates significant employment in the region.
Significance: Important for manpower planning and socio-economic analysis.

📊 8. Cost Distribution (%)



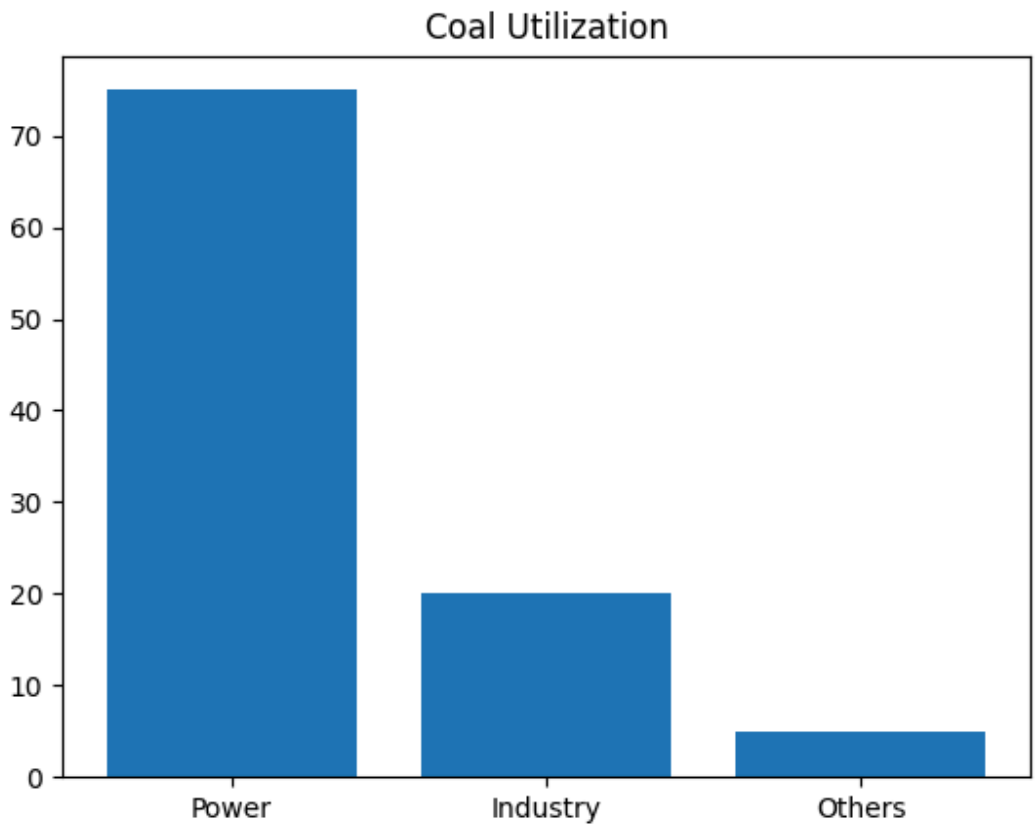
Shows breakup of mining costs into major components.
Overburden removal accounts for the largest share.
Significance: Helps in cost optimization and improving operational efficiency.

📊 9. Coal Dispatch Mode



Represents transportation of coal via rail and road.
Rail transport dominates due to bulk handling efficiency.
Significance: Important for logistics planning and reducing transportation cost.

📊 10. Coal Utilization



Shows how coal is used across power, industry, and other sectors.
Power sector consumes the majority of coal.
Significance: Highlights the role of Ib Valley in India's energy generation.